

5 The eye is a sense organ specialised for detecting light stimuli.

(a) Pupils of the eye react to light.

The effects of different light intensities from two light sources have been investigated.

The table shows the results of this investigation.

Light intensity / a.u. per m ²	Mean pupil diameter / mm	
	Incandescent light	LED light
0	25.5	25.5
75	24.0	22.0
150	24.0	18.5
200	23.0	17.5
300	21.5	16.0
650	17.5	12.0
1400	16.0	9.5

(i) Describe the relationship between light intensity and pupil diameter.

(2)

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- (ii) Calculate the amount of incandescent light that can pass through the pupil when the light intensity is 1400 a.u. per m^2 .

Use a value of pi (π) = 3.14

(3)

Answer

- (iii) Describe how light entering the eye causes the pupil to respond.

(4)

This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

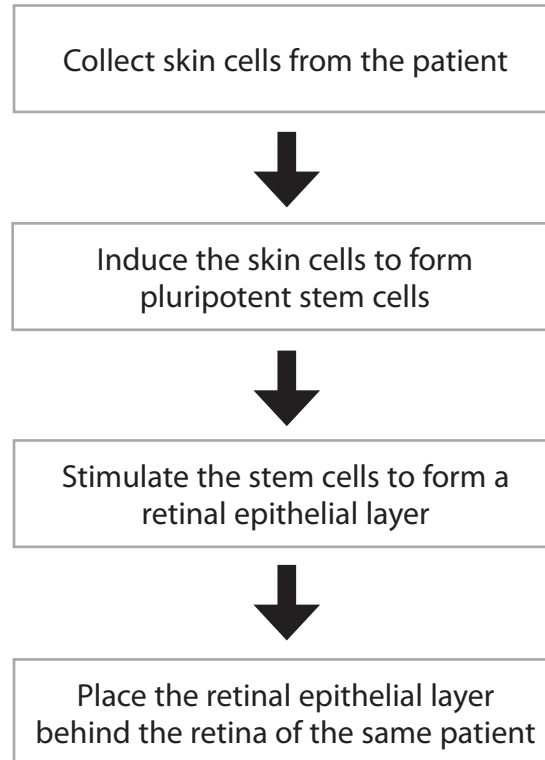
(b) Age-related macular degeneration (AMD) is a common cause of blindness in older people.

In AMD the epithelial layer behind the retina is damaged.

A new treatment for AMD uses induced pluripotent stem cells. These cells are grown in a culture to produce a retinal epithelial layer.

This retinal epithelial layer can then be used to replace the damaged layer of cells.

The procedure is outlined in the flow chart.



(i) State what is meant by the term **pluripotent stem cell**.

(2)

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- (ii) The epithelial layer is checked to ensure that there are no stem cells present before it is placed in the eye of the patient.

Suggest why this check is carried out.

(2)

(Total for Question 5 = 13 marks)